

PAPER_967: NS Phonon Tidal Deformability Correction

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Abstract

The SCm phonon correction to neutron star tidal deformability Λ bridges the gap between GR-only predictions and LIGO/Virgo observations for GW190425. The correction $\delta\Lambda = F_{UBi}/F_U \cdot \Phi_{1.25\text{THz}} \cdot 0.1$ yields a 10% maximal shift in Λ , consistent with the mass-gap nature of the lighter component.

1. Tidal Deformability Correction

$$\Lambda_{\text{UQFF}} = \Lambda_{\text{GR}} \cdot (1 + \delta\Lambda_{\text{phonon}})$$

$$\delta\Lambda_{\text{phonon}} = \frac{F_{UBi}}{F_U} \cdot \Phi_{1.25\text{THz}}(\omega_{\text{SCm}}, \Gamma) \cdot 0.1$$

2. Phonon Occupation

$$\Phi_{1.25\text{THz}}(\omega, \Gamma) = \exp\left(-\frac{(\omega - \omega_{\text{SCm}})^2}{2\Gamma^2}\right) \cdot S_{26}$$

References

1. LIGO/Virgo - GW190425 (2020)
2. Murphy, D.T. - Star Magic UQFF Framework (2024-2026)